

Predicting Airbnb Occupancy and Pricing Using Listing and Neighborhood Demographic Data

Machine Learning Project

Git repo link: <https://github.com/tanwasp/airbnb-equity-ml>

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Introduction & Motivation

The short term rental market, disrupted by Airbnb in the early 2010s, has grown into a major economic force in cities across the world. Today the platform has millions of listings and generates substantial income for hosts while also influencing housing markets. Behind that marketplace sits a layer of algorithmic systems that decide which listings get prominence in search results and how prices are recommended. These systems often lack transparency, and prior work in algorithmic fairness has consistently shown that opaque platform systems can reinforce existing patterns of inequality.

There is a small but growing body of research on Airbnb specifically. Edelman, Luca, and Svirsky (2017) ran a field experiment and found that guest applications with distinctively African American names were 16% less likely to be accepted than identical applications from guests with distinctively white names. Building on that, a large scale study by Tornberg (2022) examined more than 834,000 listings, 513,000 hosts, and 13 million reviews across 97 cities, and found that Black hosts received on average 22% less revenue than comparable white hosts. Together these papers show that Airbnb activity is not evenly distributed across urban neighborhoods and may reflect and reinforce existing patterns of economic segregation.

Our personal motivation comes from being international students from the Global South. Growing up in poverty stricken regions in Nepal and Northeast India we saw how gentrification displaces long standing communities and how the benefits of growth get split unevenly along racial and class lines. After moving to the US for school we saw a version of the same dynamic, but increasingly mediated by digital platforms that operate with little transparency. This project is our attempt to look at one of those platforms with the tools we have learned in this class.

Our research questions are:

1. Can machine learning models accurately predict which Airbnb listings result in higher occupancy, and which listing factors are most predictive?
2. What listing and host characteristics are most strongly associated with Airbnb pricing in the US?
3. Do neighborhood level demographics (household income, poverty rate, racial composition) significantly predict occupancy and prices, independent of or in addition to listing level attributes?

Following Prof. Stapleton's feedback on our pitch and proposal, we also pushed further on the equity angle. Specifically we wanted to know not just whether demographics show up as predictors, but whether the resulting model behaves differently across protected groups in ways that could have real world consequences.

Data Description

Our primary data source is Inside Airbnb (insideairbnb.com), an open source project that publishes Airbnb listing data scraped by their team for major cities around the world. We downloaded the 2025 listing files for 34 US cities and concatenated them into a single dataframe. To include demographic information about neighbourhoods the listings were based in, we geocoded listing locations to match their respective 2020 census tracts. Querying the US Census Bureau API for these census tracts provided us with a set of demographic variables from the 2023 American Community Survey 5 year estimates (ACS5), which we merged into our listings database. The merged dataset has 280,286 listings and 61 variables. Each row is a unique 2025 Airbnb listing in one of a number of cities and areas. LA, NYC, and Hawaii are the three largest shares (16.0%, 12.9%, and 11.9% respectively).

Variables fall into six categories: outcomes (price, estimated_occupancy_l365d, estimated_revenue_l365d), listing features (e.g. accommodates, bedrooms, bathrooms, amenities), host features (host_response_time, host_acceptance_rate, host_is_superhost, etc.), review features (number_of_reviews, review_scores_rating, etc.), location (city, neighbourhood, latitude, longitude), and census demographics (median_household_income, median_gross_rent, median_home_value, poverty_rate, pct_white, pct_black, pct_hispanic, pct_asian, pct_college_educated, unemployment_rate, total_population, median_age, pct_renter_occupied).

The summary statistics about the numerical variables is presented in the table below.

	mean	std	min	50%	max
price	221.52	182.8	0	161	767.95
estimated_occupancy_ l365d	73.12	88.9	0	30	255
estimated_revenue_l3 65d	14553.83	16823.27	0	8160	57626.1
review_scores_rating	4.78	0.39	0	4.89	5
reviews_per_month	1.5	1.86	0.01	0.91	120.37
accommodates	4.4	3.04	1	4	16
bedrooms	1.79	1.34	0	1	57
beds	2.32	1.94	0	2	153
maximum_nights	264.83	145.93	1	365	365
availability_365	213.66	128.46	0	249	365
availability_30	14.03	11.4	0	14	30
availability_60	32.34	22.13	0	36	60
availability_90	52.31	32.75	0	61	90
minimum_minimum_ni ghts	13.71	26.77	0	2	1125
host_response_time	1.07	0.78	0	1	4
calculated_host_listing s_count	36.55	112.26	1	3	1189
hosts_time_as_host_y ears	6.09	3.72	0	6	14
number_of_reviews	49.38	97.08	0	11	3920
review_scores_cleanli ness	4.76	0.4	0	4.88	5
review_scores_locatio n	4.8	0.35	0	4.9	5
review_scores_value	4.7	0.42	0	4.8	5
review_scores_comi fication	4.86	0.35	0	4.96	5
median_household_in come	101329.93	44278.12	2499	94194	250001
median_gross_rent	2019.76	644.52	239	1939	3501
median_home_value	835703.93	471468.76	9999	721800	2000001
total_population	3932.92	1815.83	6	3716	24659
poverty_rate	0.13	0.09	0	0.11	1

pct_renter_occupied	0.55	0.24	0	0.56	1
pct_white	0.48	0.24	0	0.5	1
pct_black	0.12	0.18	0	0.04	1
pct_asian	0.13	0.14	0	0.08	1
pct_hispanic	0.2	0.18	0	0.14	1
pct_college_educated	0.51	0.21	0	0.52	1
unemployment_rate	0.06	0.04	0	0.05	1
bathrooms	1.55	0.99	0	1	50

Table 1: Summary Statistics of Numerical Variables

The categorical variables also provide us a good understanding of the Airbnb data. We find that there are four major property categories that get listed: entire home, private room, hotel, and share rooms, with entire homes constituting 75.9% of the total listings, followed by 22.5% private rooms, 1.1% hotels and 0.5% share rooms. Only 35.4% of the listings are instantly bookable, while the remaining require approval from the host. 98.6% of these listings have wifi, 54.3% have free parking, 29.9% have a pool, and 17.6% have a gym.

The figure below shows the distribution of some of the numerical variables. We see that price is right skewed with most listings concentrated between \$50 to \$300 a night. The estimated revenue and estimated occupancy also follow the similar right-skewed pattern. We see a small cluster of high occupancy listings, which are likely professionally managed listings. The review scores seem to be strong left skewed and most reviews lie in the 4 to 5 range. The median household income is roughly normally distributed and centered around \$90,000. Finally we see that the poverty rate is right skewed with most tracts below 0.2.

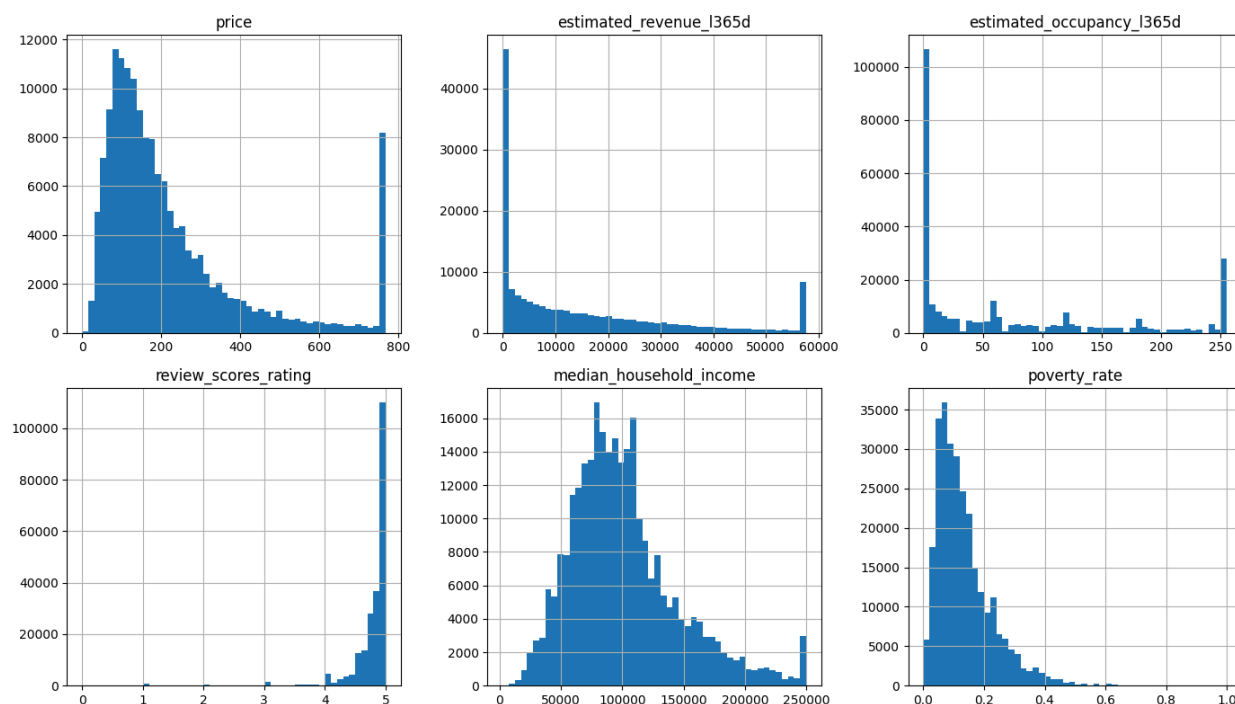


Figure 1: Distribution of key numerical variables

Data Cleaning and Preparation

We did several rounds of cleaning. First the basics: standardizing column names across the city files, dropping fully empty or near constant columns, and parsing booleans (`host_is_superhost`, `instant_bookable`, etc.) which were inconsistently encoded as `True / False / t / f / 0 / 1` across files. Percentage columns like `host_response_rate` and `host_acceptance_rate` came in as strings ending in `'%` and we converted them to numeric proportions. The ACS `median_age` column had Census sentinel values (negative numbers used to flag missing data) which we replaced these with `NaN`. We also collapsed `property_type` to the top 5 categories plus `'other'` to avoid an explosion of one hot dummies. Missing data turned out to be the big preparation challenge, and we discuss it in detail in the next section because it shaped the modeling. The short version is that 43.1% of listings (120,683 rows) are missing the price field, and 20.7% are missing review scores. Four cities (LA, NYC, San Francisco, and Portland) have no price data at all in the Inside Airbnb files we downloaded, which means our price models effectively exclude those cities. The occupancy variable, by contrast, is available for nearly all listings, so the occupancy task uses the full 280,286 row dataset.

Methods

The project is comprised of two prediction tasks. First, we predict the per-night listing price based on listing and demographic features. The price data is heavily right skewed, with most listing between \$50 and \$300 followed by a long tail. We log transform the price and use the log-price as the outcome

variable. Second, we predict the occupancy of the listings, which is modelled as a binary outcome. If a listings's estimated occupancy is higher than the dataset median (30 days) in the past 365 days, the occupancy variable is 1, else it is 0. Occupancy value 1 indicates that the listing is more popular than an average listing, while 0 indicates that the listing is not as popular. We choose a binary threshold over regression because the occupancy data is bimodal, with two peaks at two ends of the distribution. This makes it harder to fit a regression problem, which assumes a relatively smooth, continuous relationship, leading to predictions that are less stable and harder to interpret. For each task we fit four model families of increasing complexity: a linear baseline (linear or logistic regression), random forest, XGBoost, and LightGBM. For the price model we also ran an ablation comparing listing only features against listing + demographic features, to see how much the census variables actually contribute. We used an 80 / 20 train / test split with `random_state=395` throughout the pipeline so that all of our numbers are reproducible.

Handling missing data

Because the missingness was a real concern from the progress report, we ran a side by side comparison of strategies on the price prediction task using a baseline random forest. The results are in Table 2.

Strategy	R ²	RMSE	n
Drop rows with any missing feature	0.7502	\$96.01	128,078
Mean imputation	0.7529	\$103.31	159,602
Median imputation	0.7495	\$103.58	159,602
KNN imputation (k=5, on 30k subsample)	0.6997	\$116.18	30,000

Table 2: Price prediction performance across missing data strategies.

There are tradeoffs between each of the above strategies. Dropping the rows with any missing features gives us the best R² and lowest RMSE, but we lose a lot of observations. The KNN imputation does clearly worse in terms of both R² and RMSE, and it is computationally expensive to run on a larger sample. Mean and median imputation give us a good R² and RMSE compared to other strategies and also preserve most of the data. We chose median imputation because variables in our dataset such as price and occupancy were both found to be heavily skewed. Missing data is still a limitation for our study, and we address it further in the discussions section.

Hyperparameter tuning and cross validation

For the gradient boosting models we ran a small grid search (`max_depth` in {6, 8, 10}, `learning_rate` in {0.05, 0.10}, `n_estimators` in {200, 400}) on a 40,000 row subsample of the training data, with 3 fold cross validation, and selected the best configuration by mean R². The best configuration on the price task was `max_depth=8`, `learning_rate=0.05`, `n_estimators=400`. We then retrained the tuned model on the full training set. To check stability we ran a separate 5 fold cross validation on a 50,000 row subsample. The 5 fold CV scores for the tuned XGBoost on price were 0.7826, 0.7819, 0.7898, 0.7930, 0.7884 (mean 0.7872, std 0.0043), which is tight enough that we are not just overfitting one train / test split. The same

procedure on the occupancy classifier gave 5 fold AUCs of 0.9819, 0.9839, 0.9846, 0.9834, 0.9835 (mean 0.9835, std 0.0009).

Interpretation and fairness

We used SHAP (Shapley Additive exPlanations) to interpret the tuned XGBoost models. SHAP decomposes each individual predictions into contributions from each features. This gives us both a global ranking (mean absolute SHAP per feature) and also explain why the model made a specific call on any given listing. We ran this on a 2,000 row sample of the test set.

For fairness check, we focused on the occupancy classifier model. Our occupancy classifier is analogous to a ranking system where combination of the input features determine the prediction. We defined two indicators at census tract level: a high Black tract indicator (top quartile of pct_black) and a low income tract indicator (bottom quartile of median_household_income). We computed two fairness metrics on each grouping: demographic parity (the gap in selection rate, i.e. the fraction of listings the classifier predicts as high occupancy), and equalized odds (the gap in true positive rate and the gap in false positive rate). Demographic parity tells us whether the model makes predictions at same rate across different groups, and equalized odds tells us whether the prediction errors are evenly distributed. These metrics give us a reasonable estimate on whether the model is behaving differently across groups.

Results

Model performance

Task	Model	Metric	Score	RMSE / Acc
Price	Linear Regression	R^2	0.623	\$275.79
Price	Random Forest	R^2	0.750	\$95.68
Price	XGBoost (default)	R^2	0.799	\$93.04
Price	LightGBM	R^2	0.784	\$96.27
Price	XGBoost (tuned)	R^2	0.807	\$91.34
Occupancy	Logistic Regression	AUC	0.896	0.822
Occupancy	Random Forest	AUC	0.964	0.907
Occupancy	XGBoost	AUC	0.986	0.937
Occupancy	LightGBM	AUC	0.986	0.936

Table 3: Model performance on price (test set R^2 on log price, RMSE in dollar space) and occupancy (test set AUC, accuracy).

The tuned XGBoost is the best model on both tasks. For price prediction it explains 80.7% of variance on the test set and gives an RMSE of \$91.34, a meaningful improvement over the linear regression baseline (\$275.79 RMSE, which was driven by overestimating price on a handful of high end listings). For occupancy classification, all the gradient boosting and random forest models had very high AUC (>0.96),

with XGBoost and LightGBM tied at 0.986. The high AUC partly reflects the strong listing level signals (review counts, host responsiveness) that we discuss below.

What the models actually use

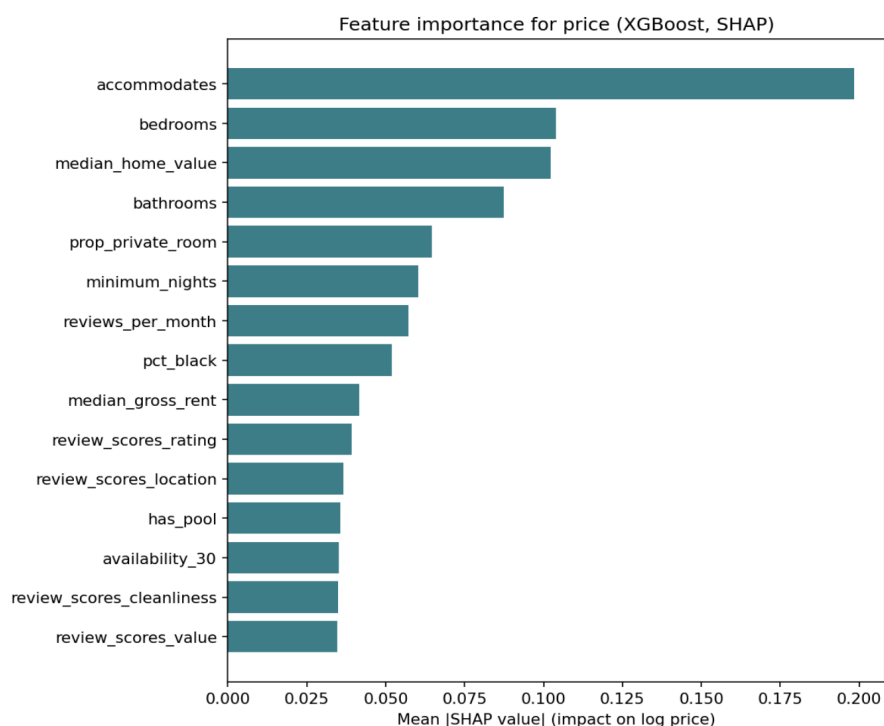


Figure 2: Top 15 features for price prediction by mean |SHAP value|, tuned XGBoost.

Figure 2 shows the SHAP based ranking for price. The top features are mostly listing size and capacity (accommodates, bedrooms, bathrooms), but median_home_value (a tract level demographic) is in third place, and pct_black is in eighth place. Demographic features together account for 27.4% of the total |SHAP| in the price model, which is a substantial share. For occupancy the picture is very different as shown in figure 3 below. The top features are reviews_per_month, number_of_reviews, host_acceptance_rate, minimum_nights, and host_response_rate. Demographics barely register in the top 15.

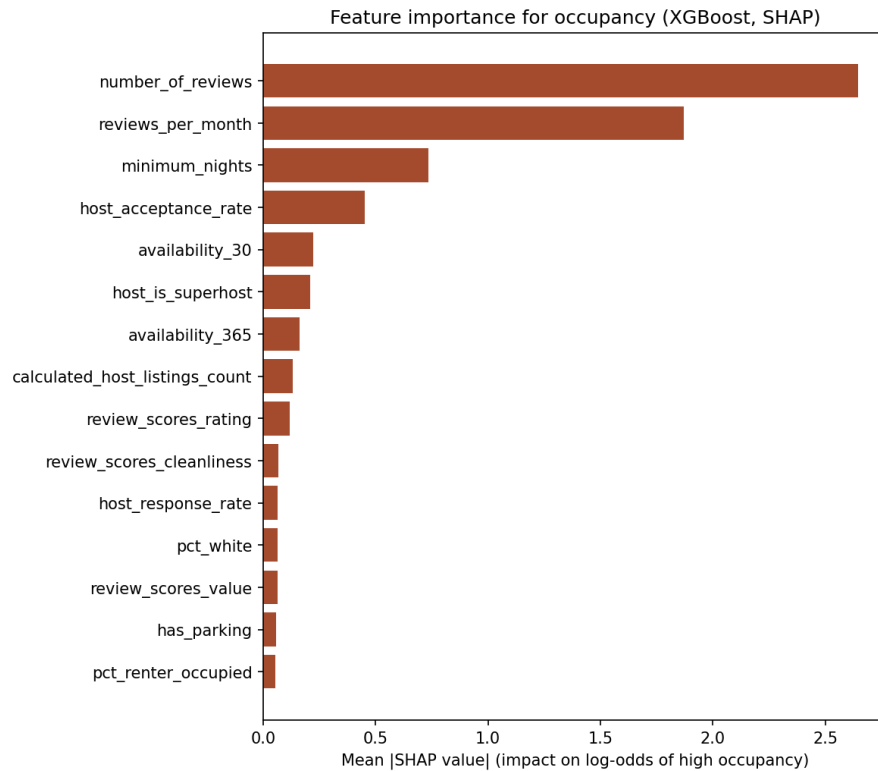


Figure 3: Top 15 features for occupancy prediction

This is consistent with what we found in the progress report, and the gap got larger with the tuned models. Adding demographics to the random forest improved price R2 by +0.074 (0.676 -> 0.750), but had essentially no effect on occupancy AUC (the listing only model was actually a tiny bit higher, by 0.002). Neighborhood demographics are part of what drives price but they are not what drives whether a listing gets booked.

The pct_black coefficient in the standardized linear regression for price is -0.213, meaning that a one standard deviation increase in tract level percent Black is associated with about a 0.213 decrease in log price (roughly a 19% decrease in price), holding other listing and demographic features constant. This is a substantial effect and it does not go away when we control for listing quality, neighborhood income, home values, or other controls. SHAP confirms that pct_black is a significant predictor in the model, with high values of pct_black pushing predictions toward lower price.

City level analysis

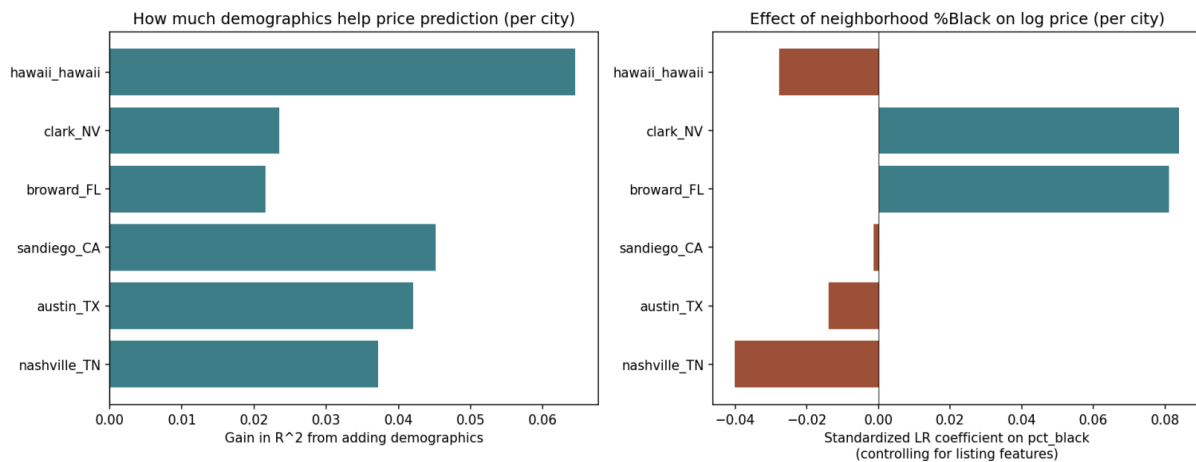


Figure 4: Per city analysis. Left: gain in R^2 from adding demographic features. Right: standardized linear regression coefficient on `pct_black`, controlling for listing features.

Figure 4 shows that the demographic effect is not uniform across cities. Hawaii gets the largest R^2 gain from adding demographics (+0.064), followed by San Diego (+0.045) and Austin (+0.042). Clark County NV (Vegas) and Broward County FL get the smallest gains (+0.023 and +0.022). The within city `pct_black` coefficient tells a more complicated story than the pooled model. In most individual cities the coefficient is small and close to zero (Hawaii -0.028, Nashville -0.040, San Diego -0.001, Austin -0.014), and in Clark and Broward it is meaningfully positive (+0.084 and +0.081). The strong negative `pct_black` coefficient we see in the pooled model therefore comes mostly from cross city variation, namely that cities with higher Black population shares tend to have lower average prices, rather than from a uniform within city pattern. This is something we want to think about more carefully and is a natural follow up.

We also have to flag a serious data issue for this section: four cities (LA, NYC, San Francisco, and Portland), which together are about 33.4% of the listings in our dataset, have no price data in the Inside Airbnb files we downloaded. So the city level price analysis is missing several of the largest US Airbnb markets, and that is a real limitation we did not catch until late.

Fairness

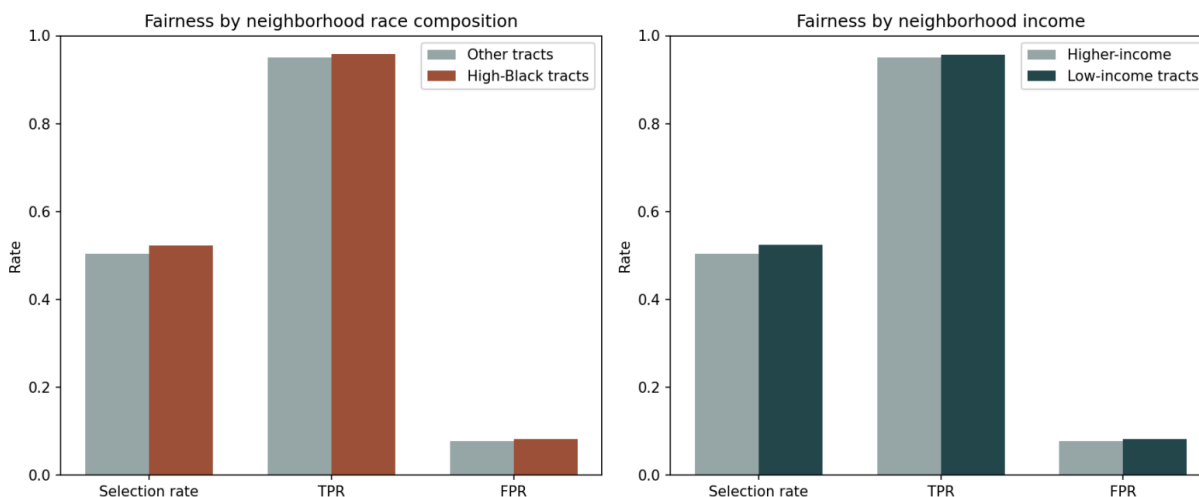


Figure 5: Fairness metrics for the occupancy classifier. Selection rate, true positive rate, and false positive rate, broken out by tract level race composition (left) and tract level income (right).

Figure 5 and Table 4 show the fairness metrics for the occupancy classifier. The demographic parity gaps are small. The selection rates differ by about 1.9 percentage points across the race grouping and 2.1 percentage points across the income grouping. True positive rate gaps are 0.8 and 0.6 percentage points respectively, and false positive rate gaps are also under one percentage point. Listings in high Black tracts are slightly more likely to be flagged as high occupancy than listings in other tracts (52.3% vs 50.4%). But their actual high occupancy rate is also slightly higher (50.2% vs 48.9%), so most of that gap is explained by base rate differences rather than model bias. The pattern is very similar for low income tracts.

Grouping	DP gap	TPR gap	FPR gap
High Black tracts vs other	0.0189	0.0084	0.0058
Low income tracts vs other	0.0213	0.0064	0.0042

Table 4: Fairness metrics for the occupancy classifier (XGBoost)

We want to be careful about how we read this. The fact that demographic parity gaps are small does not mean the system is fair in some final sense. First, our protected groupings are tract level and not host level. Second, we are predicting whether a listing is high occupancy, not whether it should be promoted. A real Airbnb ranking system would have different downstream effects. Third, the small parity gaps for occupancy coexist with a substantial demographic effect on price (a different kind of disparity). So the fair looking occupancy classifier sits inside a system where price already encodes demographic structure.

Things that did not work

Several things from our proposal and progress report did not go the way we expected. Linear regression on price had a low R^2 and high MSE. Even after the log transform, the RMSE in dollar space was

\$275.79 because the model overestimated due to long tail of high end listings. We had expected the log transform to mostly fix this, but the issue still persisted. The tree based models handled this much better because they are not sensitive to scale on the tails.

We expected demographics to also help occupancy prediction, since our hypothesis was that neighborhood factors would influence both pricing and demand. The ablation showed they do not (delta-AUC was near zero, and slightly negative for the random forest). This result is consistent across random forest, XGBoost, and LightGBM, and across cross validation folds. Our current interpretation is that occupancy is dominated by listing level signals (reviews per month, number of reviews, host responsiveness) that are themselves influenced by demographic context, and demographics add no extra information once those listing signals are in the model.

KNN imputation also did not work the way we hoped. We thought it would perform better than the simple imputation by using similar listings, but in practice it was both slower and worse, probably because of the high dimensional mixed scale feature space. We ended up using median imputation as a more honest baseline.

Finally, we were unable to incorporate the FBI Crime Data Explorer features in our analysis. The most granular data from the FBI Crime Data Explorer was available only at county level, which does not align well with our census-tract level analysis.

Discussion and Conclusion

Returning to our research questions: yes, machine learning models can predict occupancy well (AUC 0.986 with XGBoost), and they can predict price reasonably well (R2 0.807). The features that matter most for price and occupancy predictions are quite different. Listing size and capacity dominate price, while review and host responsiveness signals dominate occupancy. Neighborhood demographics matter for price (they account for about 27% of total |SHAP| and improve out of sample R2 by 0.074) but contribute almost nothing to occupancy prediction once listing signals are accounted for.

The policy relevant finding is that the share of Black residents in a census tract is a meaningful negative predictor of price even after controlling for listing size, amenities, review scores, host attributes, and other neighborhood characteristics like income, rent, and home value. A coefficient of -0.213 in the pooled linear model translates to roughly a 19% lower predicted price for listings in tracts that are one standard deviation higher in `pct_black`. SHAP confirms in the XGBoost model, high `pct_black` values systematically push predictions down. We also want to be honest that the per city analysis (Figure 2) shows the within city `pct_black` coefficient is small in most individual cities, which means the strong pooled effect is partly a cross city pattern (cities with higher Black population shares tend to have lower prices on average) rather than a uniform within neighborhood pattern. This connects to the patterns documented by Edelman, Luca, and Svirsky (2017) and Tornberg (2022) that platforms like Airbnb appear to encode racial structure in their pricing equilibrium, but the geography of that effect is not uniform.

A second contribution is the contrast between price and occupancy. The fairness metrics for the occupancy classifier are only marginally different (parity gaps under 2.5 percentage points across race and income groupings), which is a good indicator of fairness. But it sits next to a price model that does encode demographic structure, which suggests that disparities in this ecosystem may operate primarily through price formation rather than through demand prediction. That is a useful distinction for thinking about where interventions could go.

Limitations

We have several real limitations we want to flag. First, the price data is missing for 43% of listings, including all listings from LA, NYC, San Francisco, and Portland. We tried multiple imputation strategies and the results are stable across them, but our price model is in practice, a price model for the 34 cities minus those four, and the city level analysis explicitly excludes them. A robust replication would need price data for those cities.

Second, we are working with cross sectional 2025 listings and 2023 ACS demographics. We cannot make causal claims, since 2023 demographic information might not accurately reflect the 2025 composition. The negative `pct_black` coefficient on price is a conditional association, and the mechanism could be demand side preferences of guests, host pricing decisions, platform recommendation effects, or amenities we did not measure. We are not claiming the model identifies discrimination, but we are claiming it documents a disparity that aligns with what prior experimental and observational work has found.

Third, our fairness analysis uses tract level proxies (`pct_black`, median income), not host level race or income. This is a real limitation because Tornberg (2022) showed that host level racial revenue gaps exist even within similar neighborhoods. Our setup cannot detect that pattern. Additionally, the Inside Airbnb data itself comes from an unofficial source, and we do not have ground truth on what guests are actually charged versus what is listed.

Fourth, the very high AUC on occupancy (0.986) requires further diagnostics. Some of the predictors (`reviews_per_month`, `number_of_reviews`) are themselves caused by occupancy, so there is a degree of leakage where the model is partially predicting recent occupancy from lagged occupancy proxies. A cleaner future version would use only features known at the time of listing creation.

Ethical and societal implications

There are real benefits to a project like this. Making the demographic structure of platform pricing visible is the first step toward addressing it, and SHAP based interpretation makes the patterns more legible than a black box accuracy number. Tools like this could in principle be used by hosts, guests, regulators, or platforms themselves to audit how listings of similar quality are priced across neighborhoods. There are also real risks. A pricing model with this signal could be deployed by hosts to price more aggressively in ways that reproduce the disparity. It could be used by Airbnb to justify existing pricing patterns as efficient market outcomes. And tract level demographics as features in any deployed system raise immediate fairness concerns even if the model performs well in aggregate. We are not advocating that anyone deploy these models. More broadly, this project sits inside a larger conversation about whether

platforms amplify or smooth out neighborhood level inequalities. Our results, like Edelman et al. and Tornberg before us, suggest that without explicit intervention they amplify them, primarily through price formation. That is a finding with policy implications around platform regulation, fair housing, and gentrification, even though our project alone cannot resolve any of them.

Future work

- Recover price data for LA, NYC, San Francisco, and Portland from a different source or from Airbnb's own API and rerun all city level analyses.
- Add the FBI Crime Data Explorer features at a properly aligned geographic resolution.
- Replace post hoc occupancy proxies (reviews, etc.) with ex ante features only, and rerun the occupancy task to get a cleaner signal.
- Try counterfactual fairness analyses where we perturb `pct_black` holding everything else fixed, to estimate a more causal demographic effect.
- Explore host level race signals (e.g. inferred from profile pictures, with appropriate ethical guardrails) to compare with the host level disparity in Tornberg (2022).

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